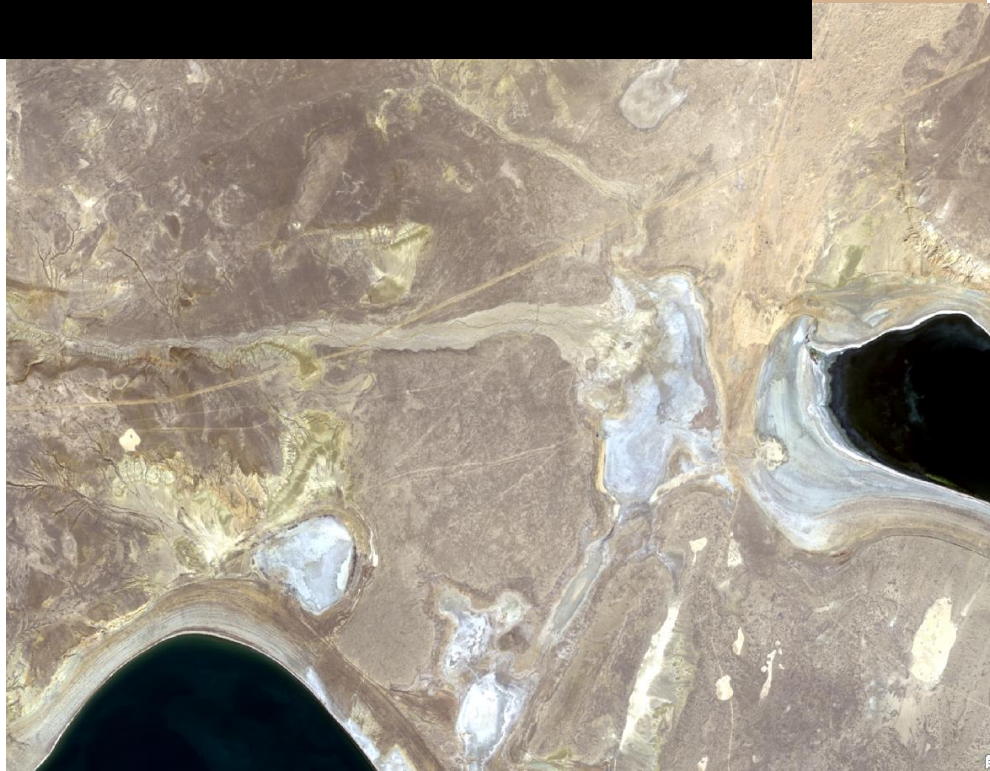


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Exploring Desertification of the Aral Sea Region: A Land Cover Analysis of Akespe, Kazakhstan



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Abstract

In this project we explore the desertification of the Aral Sea region in southwestern Kazakhstan and in particular the village of Akespe. Akespe has few remaining habitants (5 families as of 2022, from 45 in 2007*). The village is slowly being buried in sand due to high winds and drying of the adjacent Aral Sea. In 2005 a modern dam was built to raise the water level and reduce these effects. In this study we use raster analysis of Landsat satellite images from 2007 and 2021 to perform land classification in the area. Satellite imagery and World Soils maps serve as references for the classification. We examine the changes in percent area of water, soils, and vegetation to see if a correlation can be made with construction of the dam. We also look at the subjectivity involved in defining informational classes and compare land classification results between different evaluators of the same image. Our analysis shows variability among evaluators in informational class assignments, water being the most consistent among evaluators and vegetation the most variable. Using the mean of all classification methods, the year 2021 showed a decrease in water and fixed sand with vegetation informational classes over the year 2007. However, the NDVI difference showed a slight increase in vegetation over the area corresponding with fixed sand.

*Verbal information from local tour guide.

Introduction

The current environmental crisis of the Aral Sea and surrounding region has its origins in the 1960's (Cretaux et al., 2013, p. 106). The area lays in what is current day Kazakhstan and Uzbekistan and was once part of the Soviet Union. During that time, the Amu Darya and Syr Darya rivers were diverted for irrigation of wheat and cotton fields (Cretaux et al., 2013, p. 106). Being an arid lake, river inflow is important to maintain lake levels because precipitation is generally low and evaporation rates constant (Cretaux et al., 2013, p. 106). This along with regional and global climate change resulted in significant drying of the sea and the devastation of the flourishing fishing industry (Cretaux et al., 2013, p. 106). Over the course of only 4 decades the sea was reduced to roughly 50% of its early 20th century size by area (Qobil & Harris, 2018). As the water dried the salinity increased along with the percentage of pesticide and agricultural chemicals (Izhitskiy et al., 2016, p. 3; Qobil & Harris, 2018). High winds (from lack of wind break) blew the tainted soil to the surrounding area, degrading soil the soil quality (Qobil & Harris, 2018). Crops were more difficult to grow and livestock more difficult to manage (Qobil & Harris, 2018). Canneries were closed and along with them the supporting infrastructure and industry disappeared (The World Bank, 2005). Gradually, people left the surrounding towns and villages as their livelihood disappeared. Some villages were

overwhelmed with the environmental changes. The village of Akespe, featured in this study, is one of them. It is slowly being buried in sand blown over from the Aral seabed(Qobil & Harris, 2018).

Encouraged by results from a primitive dam built with earthen materials, the World Bank embarked on a project to build a modern dam spanning 8 miles to restore the Northern portion of the Aral Sea. (The World Bank, 2005). The southern, and larger portion of the sea, which lay mostly in Uzbekistan would be sacrificed and destined to become desert. Kokaral Dam was completed in 2005(The World Bank, 2005).

This study will not only include the percentage change of land classifications in the area between 2007-2021, but also attempt to draw a correlation (if any) with the building of the Dam.



Figure 1: Locator Map showing Kazakhstan Country outline and Northern Aral Sea Region .

Methods

Data Collection:

The satellite images used in this analysis were downloaded from <https://earthexplorer.usgs.gov/>. We chose two Level-1 GeoTIFF data products from Landsat Collection 1, Level-1: Landsat 5 LT05_L1TP_161027_20070826_20161111_01_T1 with seven bands; and Landsat 8 LC08_L1TP_161027_20210901_20210910_01_T1 with eleven bands. 2007 images from Landsat 7 were not used because of poor image quality. We also accessed the World Reference Base for Soil Resources 2021 from Living Atlas on ArcGIS online at <https://www.arcgis.com/home/item.html?id=44b8b4b7c21f4ee88b42ec6f35906af2>.

Landsat Product Identifier	Image Date	Bands	Spatial Resolution	WRS Path & Row
LT05_L1TP_161027_20070826_20161111_01_T1	Aug 26, 2007	7	30m	161, 27
LC08_L1TP_161027_20210901_20210910_01_T1	Sep 1, 2021	11	30m	161, 27

Table 1: Landsat satellite Level-1 GeoTIFF data product descriptions.

Preprocessing:

After downloading the GeoTIFF data products from USGS Earth Explorer, composite images including all spectral bands were made for each year in ArcGIS Pro. This was done using the Composite Bands tool from the geoprocessing pane. Background values were removed for each composite image and the Red, Green, Blue bands were changed to 3,2,1 for the 2007 composite and 4,3,2 for the 2021 composite to appear natural in color.

Next the composite images and World Reference Base for Soil Resources image were clipped by exporting the rasters to the same shapefile boundary layer surrounding the region of interest of Akеспе, Kazakhstan. The boundary layer was approximately 16miles x 16miles square. The clipped images were inspected to ensure the spatial resolutions and coordinate systems were the same at 30m and WGS 1984 UTM 41N.

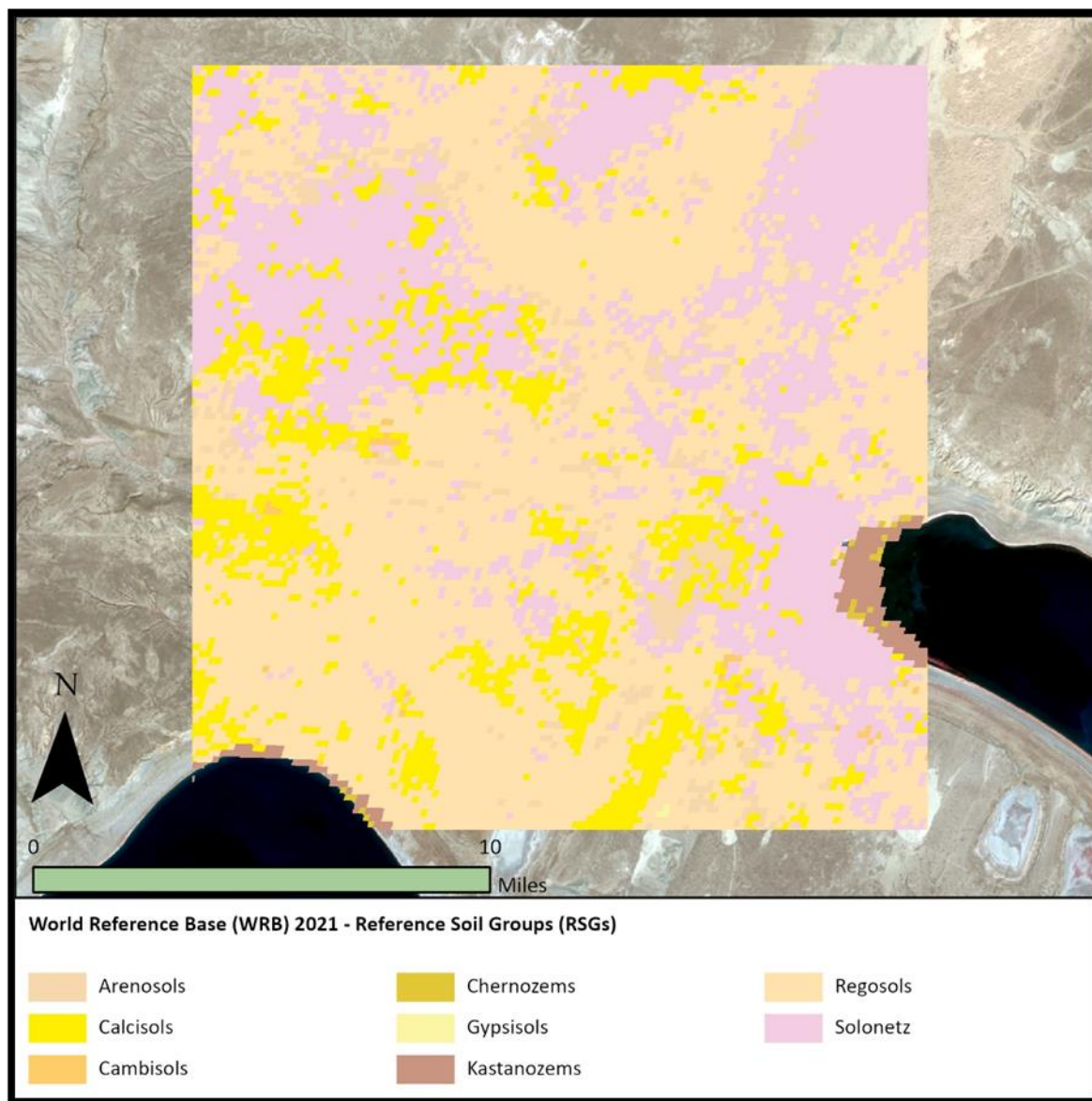


Figure 2: Percent cover of 2021 WRB Reference Soil Groups of clipped composite image used to help define informational classes.

Analysis:

For our land cover analysis in ArcGIS Pro we used the Iso Cluster Unsupervised Classification tool with 25 classes, then reclassified the image manually using the Reclassify tool into the informational classes shown in Table 2. Informational classes were decided on by looking at the World Reference Base soil groups in the area and the informational classes used in Dukhovny et. Al's paper *GIS for Sustainable Water Development in Central Asia*. A description of soil types produced by the Food & Agricultural Organization (FAO) of the United Nations was also used to decide on informational classes. Each composite clipped image was reclassified twice by two different evaluators. In addition, an attempt at a control was made for each composite clip by performing an unsupervised

classification with only 6 classes and assigning the informational class that appeared most likely. Spectral indexes like false color infrared were used in the reclassification process to help define the vegetation classes.

Informational classes	Description
Open Water	visible surface water
Fixed Sand	stable sand that does not erode easily
Fixed Sand with Vegetation	stable sand held in place by vegetation
Loose sand	unstable sand, wind deposited sand, sand with features of erosion
Alkali with vegetation	soils rich in Na, Mg, or Calcium, grey in color with vegetation
Alkali	soils rich in Na, Mg, or Calcium, grey in color

Table 2: Description of informational classes used for reclassification of unsupervised classification image.

After reclassification, percent area, and area in hectares fields were added to each reclassified clip for comparison.

In addition to unsupervised classification, an NDVI of each composite clip was made using raster functions in ArcGIS Pro. The raster calculator was used to take the NDVI difference subtracting the 2007 NDVI clip from the 2021 NDVI clip. The reclassify tool was used with classify to set the number of classes to five: 1. Significant loss (-0.57 to -0.3); 2. Loss (-0.3 to -0.03); 3. Little to No change (-0.03 to 0.09); 4. Gain (0.09 to 0.2), and 5. Significant gain (0.2 – 0.66).

Results

Results show variability in the assignment of informational classes by evaluator and control. The informational class of water was the most consistent, decreasing in the 2021 image among all classifiers. In both years the vegetation classes and fixed sand classes had the most disagreement among evaluators. Using the mean of the three classifying methods, water and fixed sand with vegetation appear to decrease by approximately 20% and 23% respectively, while other informational classes showed an increase. Alkali with vegetation showed the greatest increase in percent cover from 2007 to 2021 at ~17% using the mean. Among classifiers, the percent change from 2007 to 2021 of the informational classes varied considerably, showing increase and decrease percent changes of the same class frequently. The NDVI difference showed the majority of land area (~67%) with little to no change, ~30% with some gain, ~2% significant gain (around shores or the North Aral Sea), <1% loss, and <1% significant loss.

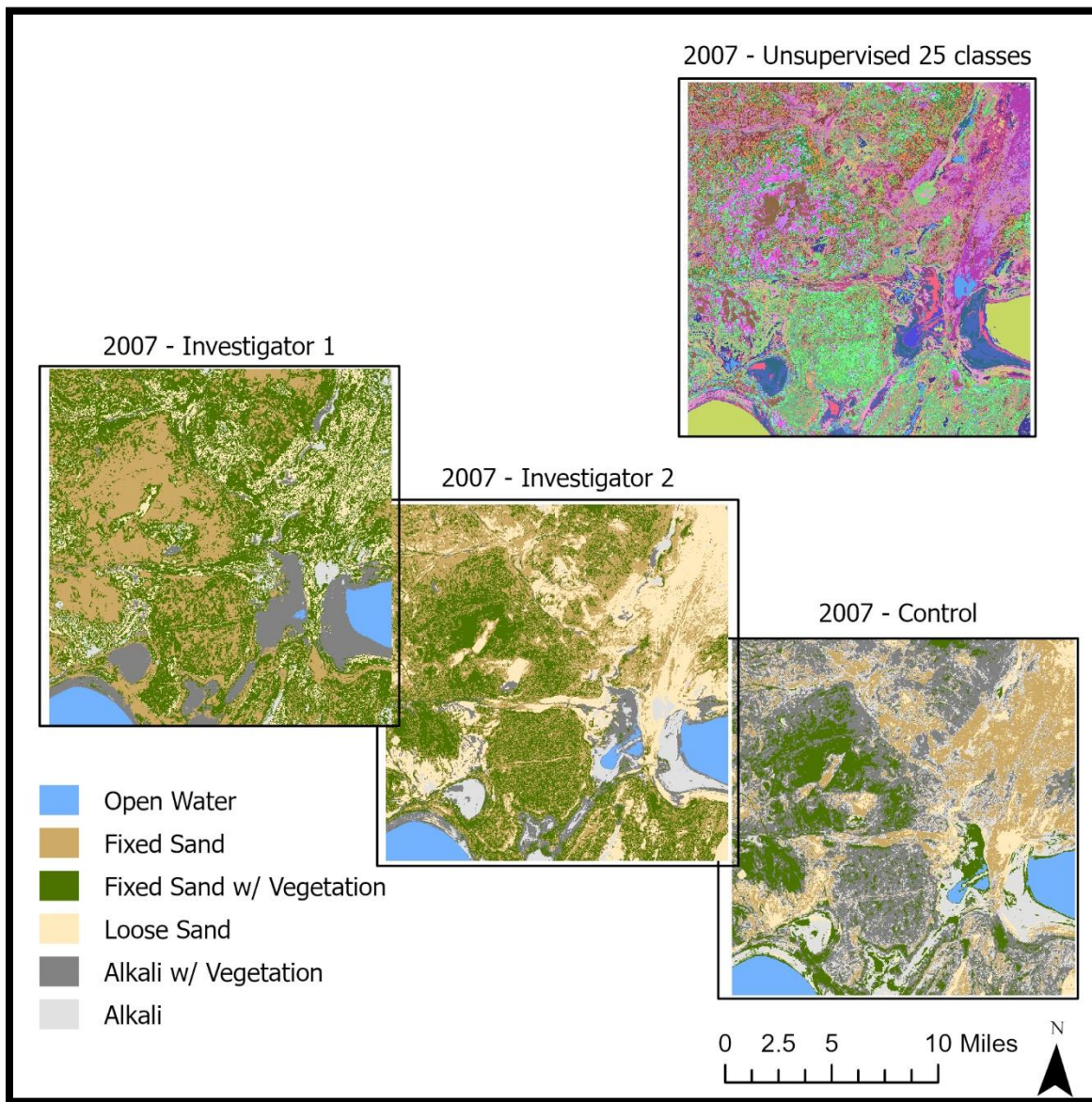


Figure 3: Percent cover of informational classes reclassified by Investigators 1, 2 and Control for Aug 26, 2007 clipped composite image.

Informational Class - 2007	% Investigator 1	% Investigator 2	% Control	St Dev	Mean: 2007
Water	3.94	4.44	4.38	0.22	4.26
Fixed Sand	26.03	33.13	19.68	5.49	26.28
Fixed Sand with Vegetation	43.94	28.85	15.67	11.55	29.49
Loose Sand	14.08	22.29	6.15	6.59	14.18
Alkali with Vegetation	9.55	6.51	31.89	11.32	15.98
Alkali	2.46	4.78	22.22	8.82	9.82

Table 3: Percent cover of informational classes by investigator and control for 2007 image.

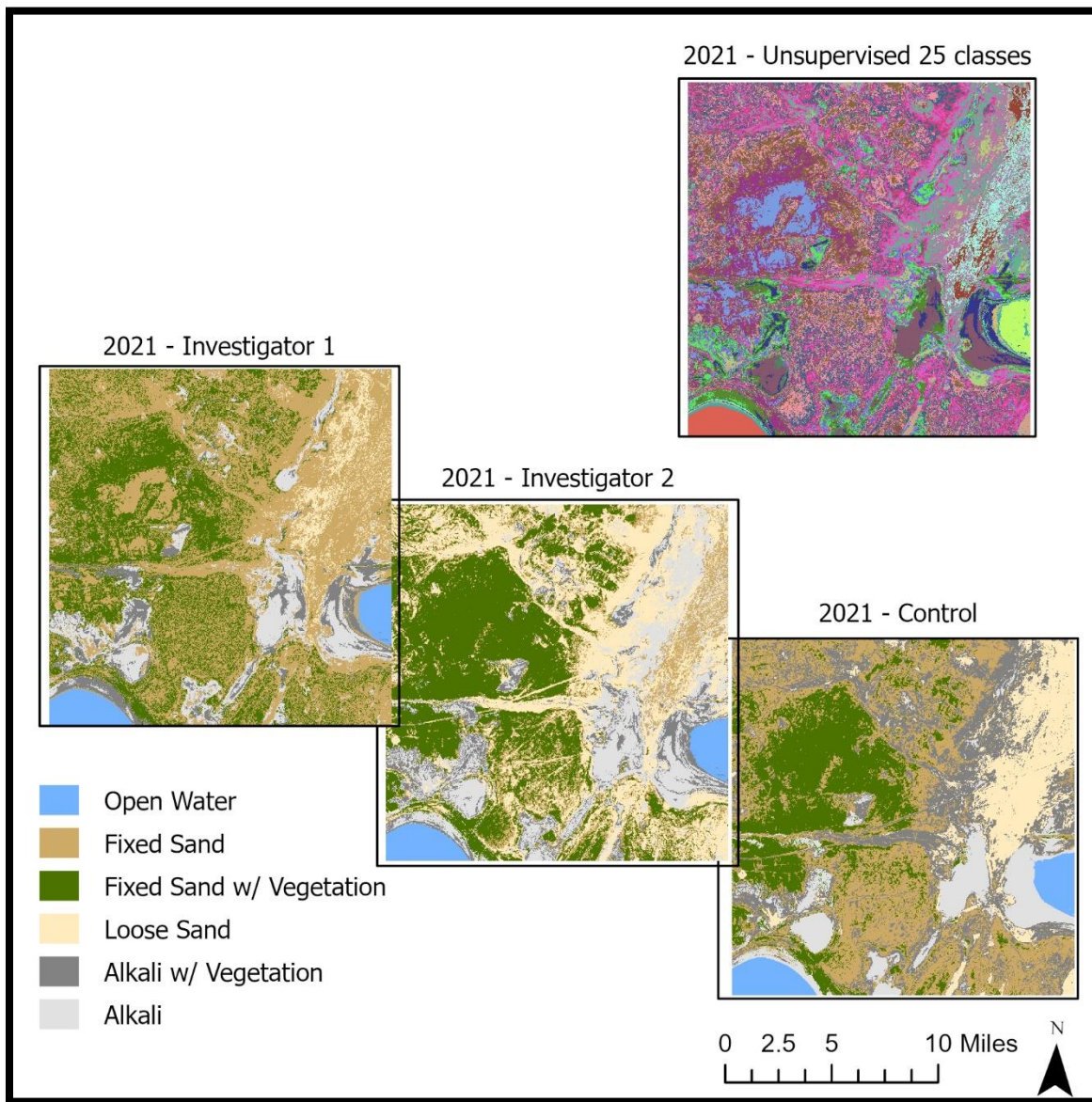


Figure 4: Percent cover of informational classes reclassified by Investigators 1, 2 and Control for Sep 1, 2021 clipped composite image.

Informational Class - 2021	% Investigator 1	% Investigator 2	% Control	St Dev	Mean: 2021
Water	3.23	3.45	3.60	0.15	3.43
Fixed Sand	48.29	15.24	20.33	14.53	27.95
Fixed Sand with Vegetation	6.19	30.24	31.90	11.75	22.78
Loose Sand	10.99	27.16	9.13	8.10	15.76
Alkali with Vegetation	27.17	6.90	22.08	8.61	18.72
Alkali	4.14	17.01	12.95	5.37	11.37

Table 4: Percent cover of informational classes by investigator and control for 2021 clipped image.

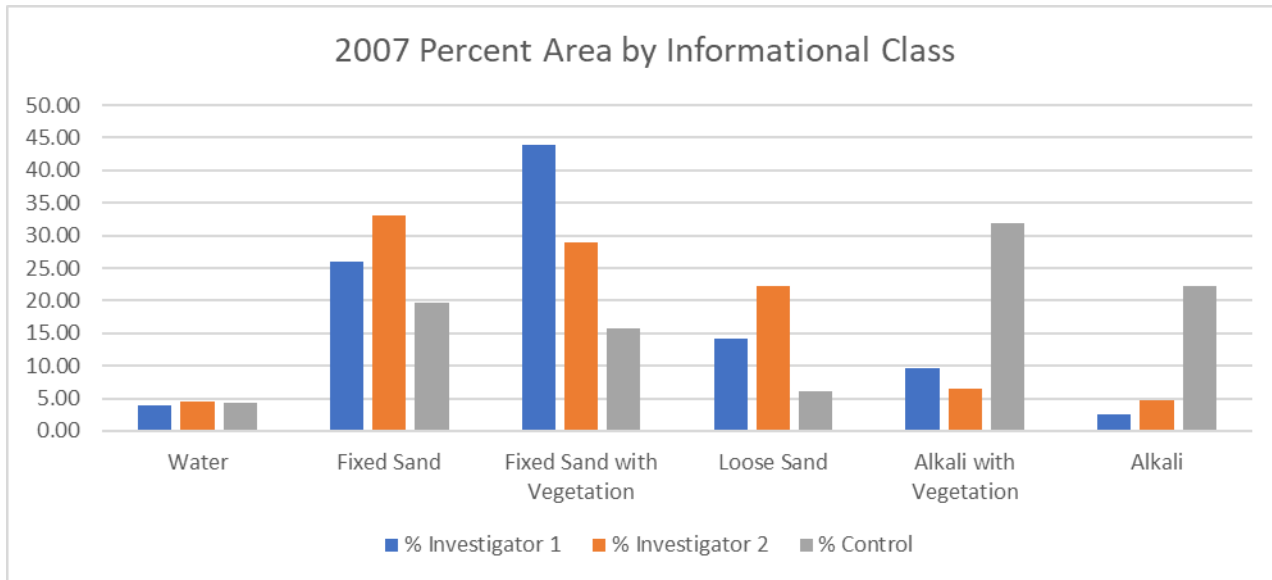


Figure 5: Percent cover of informational classes reclassified by Investigators 1, 2 and Control for Aug 26, 2007 clipped composite image.

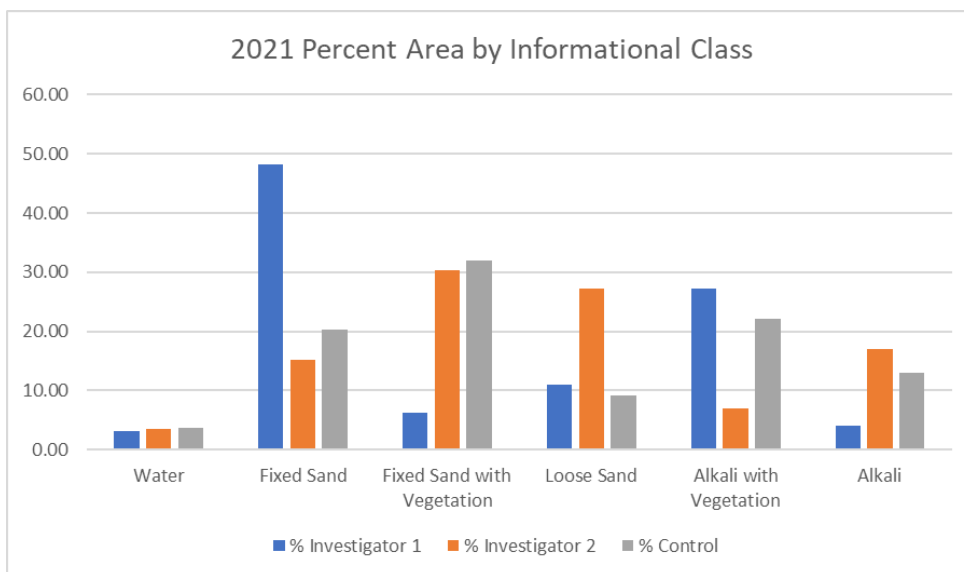


Figure 6: Percent cover of informational classes with average value for each class for year 2021.

% Change 2007 to 2021	Invst 1 % Change	Invst 2% Change	Control % Change
Water	-18.10228803	-22.37475271	-17.8950113
Fixed Sand	85.52106021	-53.98927734	3.29584803
Fixed Sand with Vegetation	-85.90834808	4.80325674	103.5595442
Loose Sand	-21.98306009	21.85583618	48.4374606
Alkali with Vegetation	184.4062201	5.962197634	-30.76587659
Alkali	68.29806253	256.1783658	-41.72027301

Table 5: Percent change in informational classes from 2007 to 2021 by evaluator

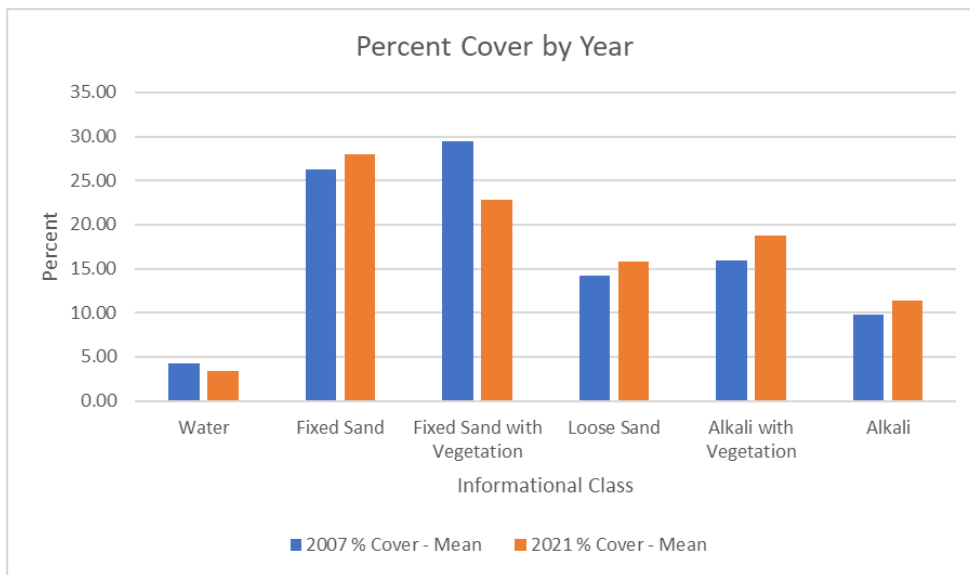


Figure 7: Percent cover of informational classes with average value for each class for years 2007 & 2021.

Informational Class	2007 % Cover - Mean	2021 % Cover - Mean	Percent Change
Water	4.26	3.43	-19.52
Fixed Sand	26.28	27.95	6.37
Fixed Sand with Vegetation	29.49	22.78	-22.76
Loose Sand	14.18	15.76	11.18
Alkali with Vegetation	15.98	18.72	17.08
Alkali	9.82	11.37	15.77

Table 6: Percent cover of informational classes with average value for each class for years 2007 & 2021.

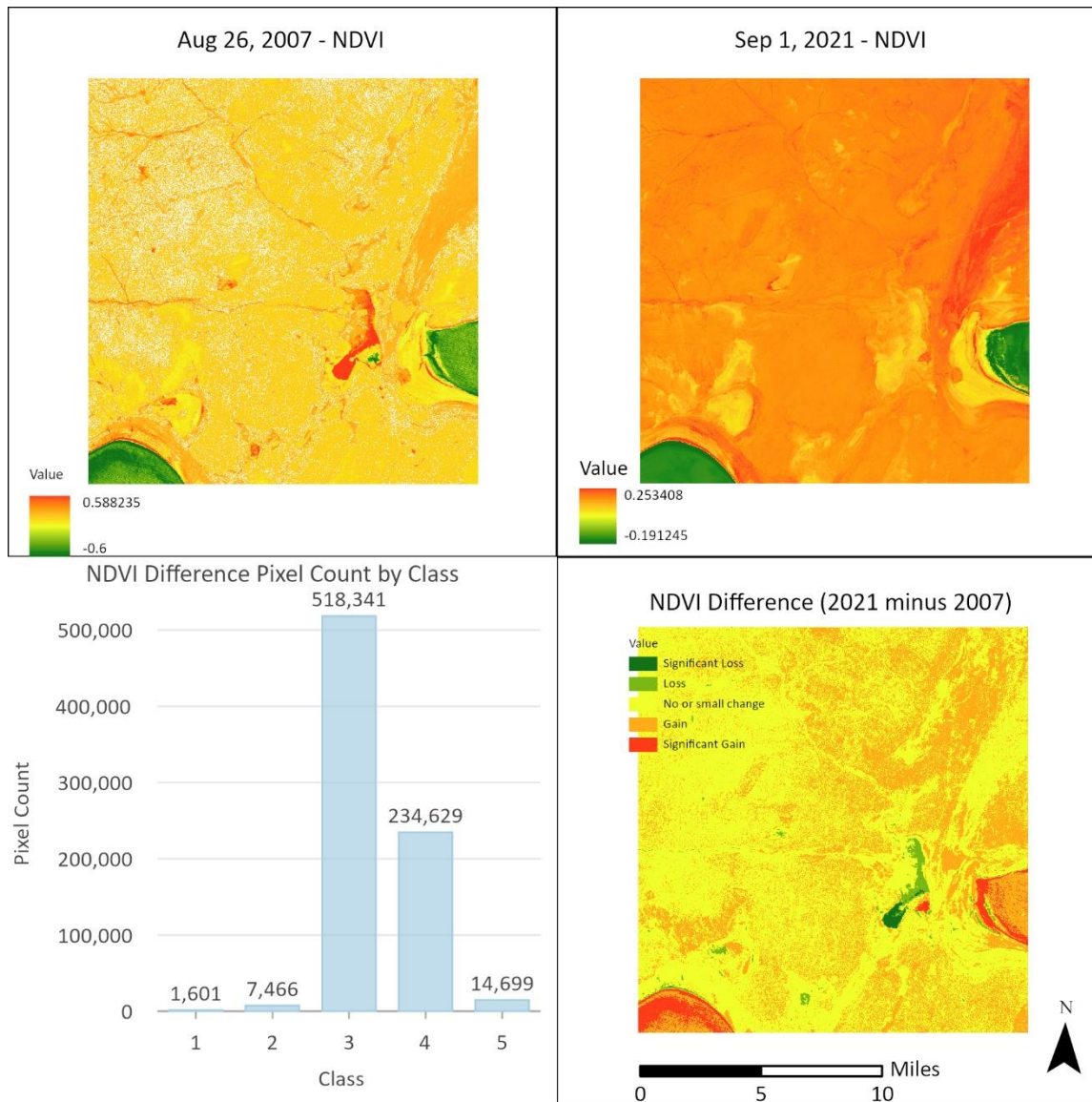


Figure 8: NDVI difference from 2007 to 2021.

Class Name	Value	Pixel Count	Percent	Area (hectares)
Significant Loss	1	1,601	0.21	144.09
Loss	2	7,466	0.96	671.94
No change	3	518,341	66.73	46,650.69
Gain	4	234,629	30.21	21,116.61
Significant Gain	5	14,699	1.89	1,322.91

Table 7 : NDVI difference from 2007 to 2021 by informational class.

Discussion

Results presented show a net increase in the areas of Fixed and Loose Sand (total sand) classes accompanied by a decrease in water and fixed sand with vegetation classes. This net increase in desertification includes many variables

outside the beneficial effect of Kokaral Dam. It appears as if the influence from variables such as global warming, extreme weather and anthropogenic effects may impact the benefit from the Dam. However, the NDVI difference does show a gain in vegetation for 2021 in some areas classified as fixed sand. This evidence supports the inference that the Kokaral Dam is having a positive affect and decreasing the effects of desertification. It is also possible the sandy terrain in the images skewed the NDVI results.

Additionally, it is presumed that the Dam at its current height has reached its maximum effectiveness in raising the sea level and would need to be rebuilt to a height 2 meters higher than present. (Utoeva, A. 2020).

Finally, this study itself involves a large component of subjectivity, as can be seen in the variation between investigators 1, 2 and the “control.” This should be taken into consideration when evaluating the results. This study shows the need for additional methods such as ground truthing classification results and using supervised classification to improve accuracy.

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